STOR 565 · FALL 2026 · FINAL PROJECT

NMF versus PCA on the Olivetti faces

Can you build a face just by adding patterns, and what does that cost compared with PCA?

Person 1 · Person 2 · Person 3 · Person 4

The question

DATA

Olivetti faces

400 photos of 40 people (10 each), 64 × 64 pixels, brightness between 0 and 1

BASELINE FROM CLASS

PCA

Components can be positive or negative, added to a mean face

NEW METHOD

NMF

Every pattern is nonnegative, and a face is built only by adding them

If a face can only be built by adding, how much accuracy do we lose compared with PCA, and are the patterns easier to read?

A fair comparison

- 320 training and 80 test images, fixed (8 + 2 per person)
- Both methods are fit on the training images only and rebuild the same 80 test images
- Error is the mean squared error per pixel (MSE), computed on unclipped reconstructions
- Both get the same number of components k (mostly k = 40)
- We never use the person labels, except to check the split

CAVEAT

Test photos are of people we trained on

Every person has 8 photos in training and 2 in test, so this says nothing about people the model has never seen.

The NMF model

$X \approx W H$, with $W \geq 0$ and $H \geq 0$

X: 320 × 4096

training images, one per row

W: 320 × 40

how much each image uses each pattern

H: 40 × 4096

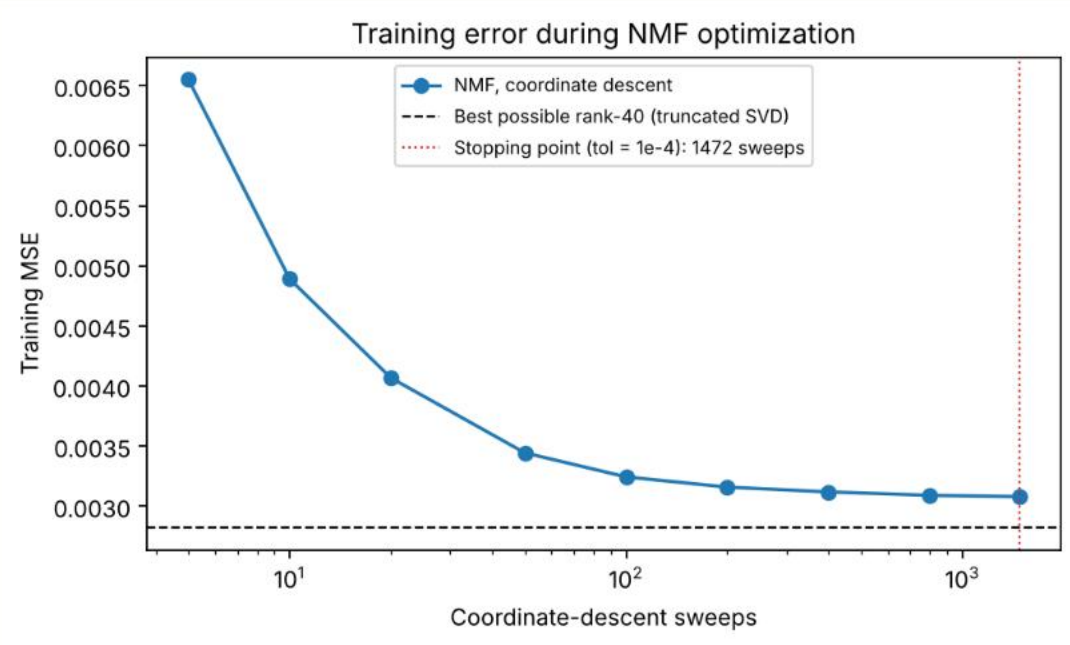
the 40 patterns; each row is one image

So a face is a weighted sum of the 40 patterns, and nothing gets subtracted. The 40 is just the size we picked. It has nothing to do with the 40 people.

We minimize $\frac{1}{2} \|X - W H\|^2$, the sum of squared pixel errors.

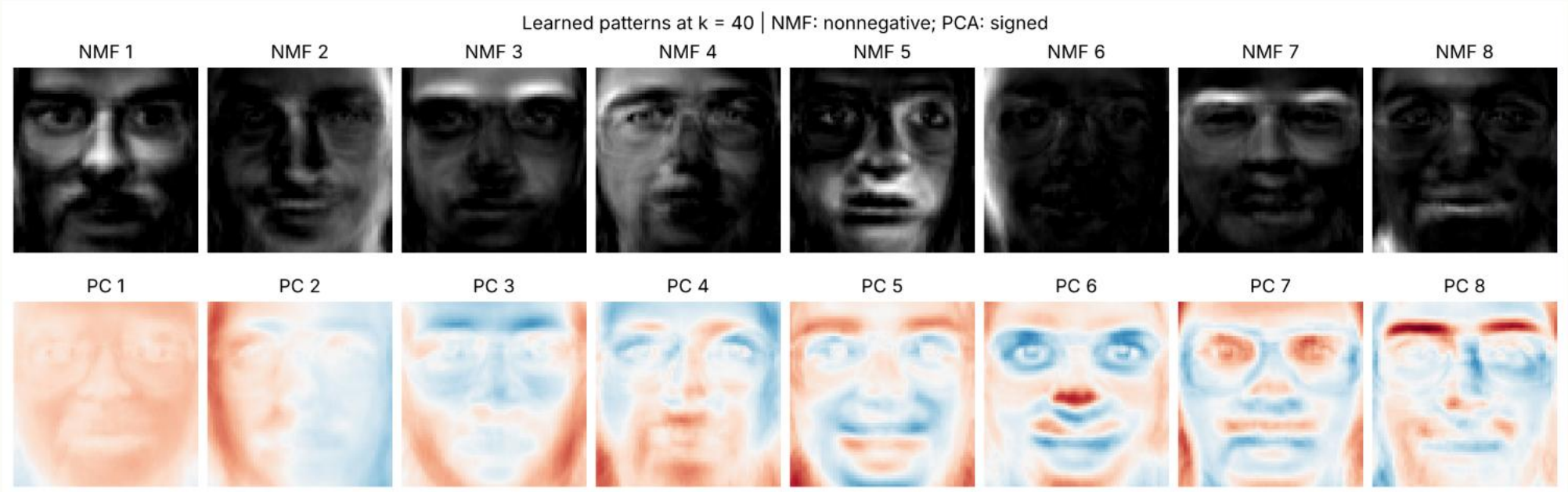
How the patterns are learned

- For fixed H the problem is convex in W, and the other way around, but it is not convex in both together
- We update one entry at a time: W[i, r] ← max(0, W[i, r] − gradient / a_r), where a_r is the sum of squares of row r of H
- Exact NMF is NP-hard, so nothing guarantees a global optimum
- For a test face we keep H fixed and solve for nonnegative weights. An exact solver gives the same test MSE, 0.00394635



The error drops quickly and then slowly. It stays above the best possible rank-40 error (dashed line).

What the NMF patterns look like



0.317 vs 0.221

Hoyer sparsity of the NMF patterns vs. PCA's absolute loadings (0 = flat, 1 = a single pixel)

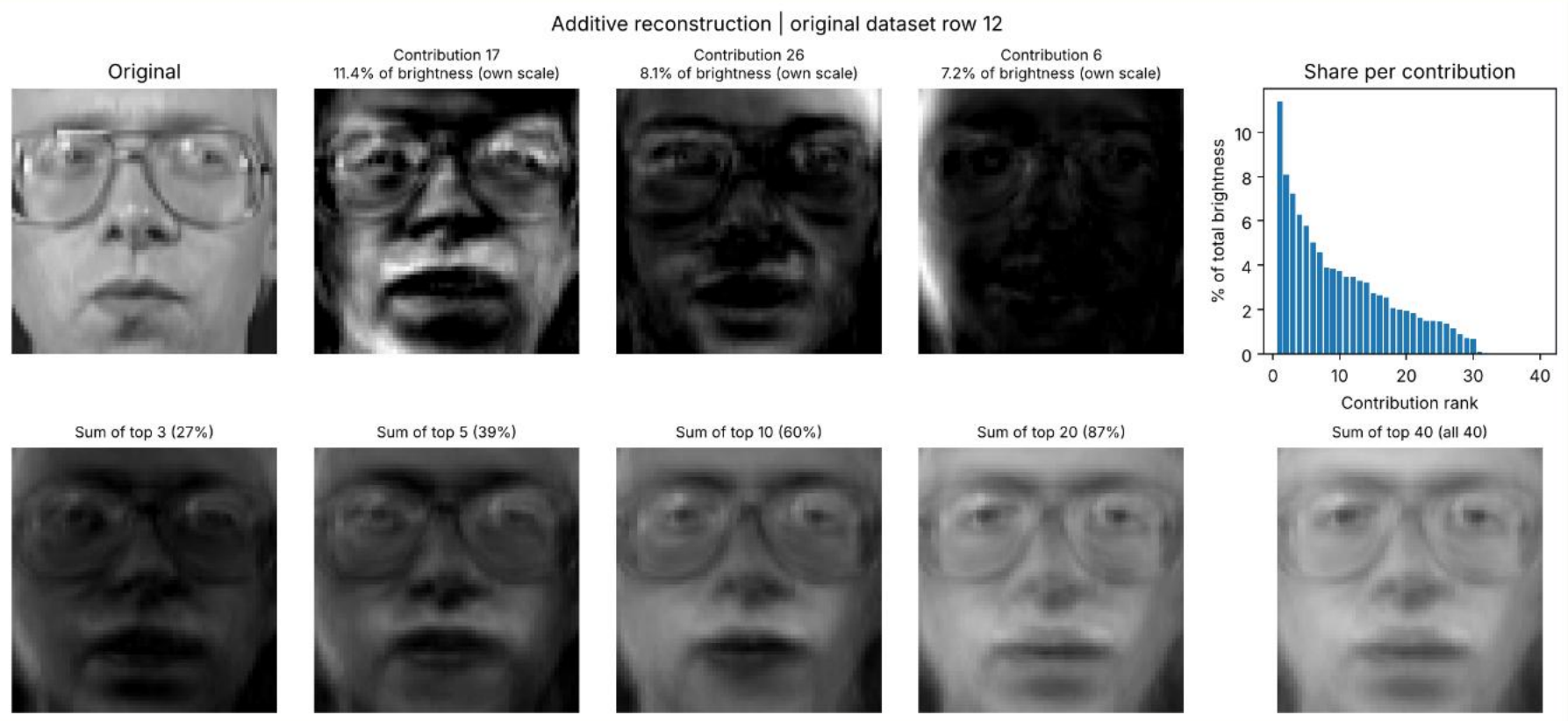
29.8 of 40

patterns used by a typical training image

Broad patterns

they overlap and look like faces; none is a lone eye, nose or mouth

One face as a sum of contributions



Test image (dataset row 12). The biggest of the 40 contributions gives 11.4% of the brightness and the top 20 give 87%, so no single pattern dominates.

Reconstruction at 40 components

Method	Test MSE	vs. PCA
PCA, 40 components + mean	0.003658	—
PCA, 39 components + mean	0.003696	+1.0%
Uncentered SVD, 40, no mean	0.003658	−0.005%
NMF, 40 components	**0.003946**	**+7.9%**

The missing mean is not the reason. PCA without a mean (uncentered SVD) does no better, and PCA with 39 components + mean, which stores as many images as NMF, still wins.

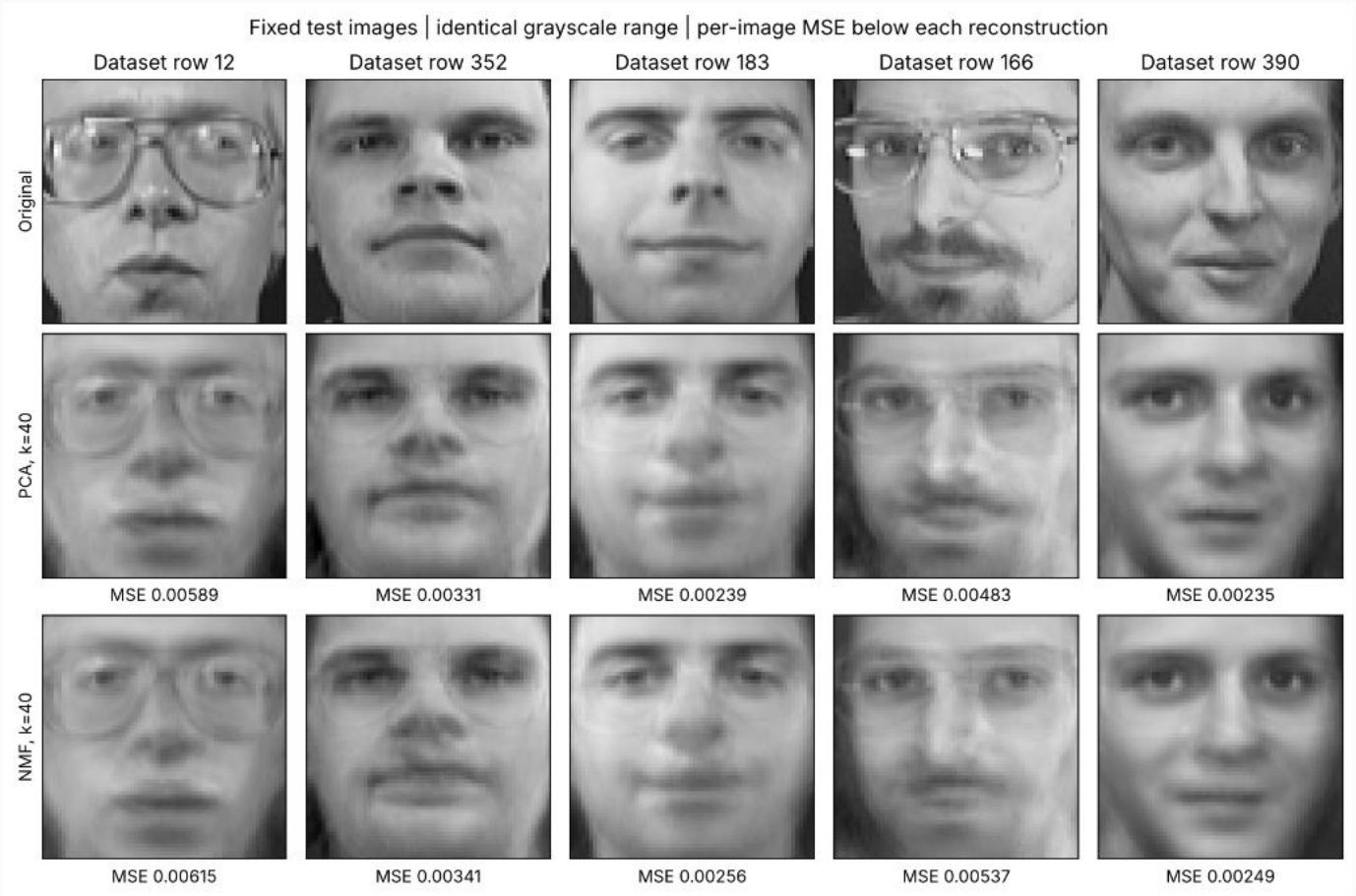
+7.9%

NMF's test MSE is higher than PCA's

74 of 80

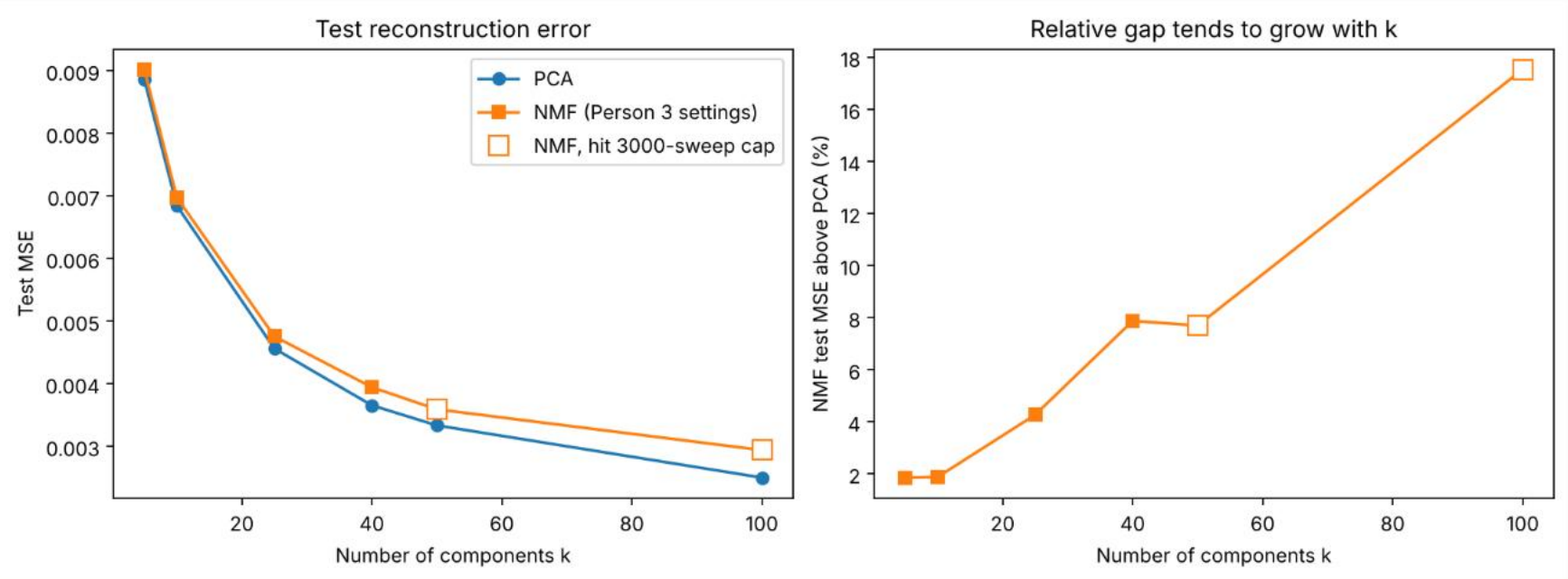
test images where PCA has the lower error

Same faces, two reconstructions



- At first glance the two look almost the same
- The biggest gap is row 166, the face with glasses and a moustache: NMF is 11.2% worse
- Lower MSE only means closer pixels. We did not test recognition.

Does the gap hold up?



+1.9% to +17.5%

NMF's error above PCA's for k = 5 to 100. The two largest k stopped at the iteration limit, so those gaps are upper bounds

+5.5% to +7.9%

gap at k = 40 with other initializations and tolerances. PCA is lower in every variant

Why is NMF worse?

Theory

By Eckart–Young, no rank-40 factorization $X \approx WH$ fits the training data better than the truncated SVD.

NMF's training MSE is **0.003079** and the SVD bound is **0.002820**, so the nonnegativity constraint costs some accuracy.

Optimization

NMF only finds a local solution, so some of the gap may come from the optimizer and not from the model.

Other starting points and a stricter tolerance move the gap between 5.5% and 7.9%, but never close it.

So the gap comes from the nonnegativity constraint and from imperfect optimization. The missing mean is not the cause.

Limits

- We used one split. The interval only covers variation across the 80 test images, not other splits or other people.
- We only measured pixel error, not recognition, clustering or anything built on top of it.
- It is one small dataset of aligned faces and one loss, squared error.
- NMF only reaches local solutions. Other optimizers or initializations might shrink the gap, which is why we report a range.
- Sparsity is measured, but whether a pattern is a meaningful part is our judgment from looking at the images.

Take-aways

1. NMF writes a face as a sum of nonnegative patterns. PCA cannot, because its components have both signs.

2. That costs accuracy: at 40 components NMF is **7.9% worse**, and PCA wins on 74 of 80 test images.

3. The mean term does not explain the gap, and the gap stays in every variant we tried. What we get for the lost accuracy is a reconstruction that is a plain sum of patterns.

Code, data script and results: github.com/MachineWJQ6657/pca-vs-nmf

Backup: likely questions

Question	Short answer
Why does NMF not win at its own objective?	The SVD is the best rank-40 fit with no constraints, so NMF's training error cannot go below it (0.003079 vs. 0.002820).
Could better optimization close the gap?	Maybe part of it. A stricter tolerance and five random starts move the gap between 5.5% and 7.9%, and PCA stays lower.
Is it the missing mean?	No. PCA without a mean gives the same test error as centered PCA, and PCA with 39 components + mean still beats NMF.
How do you encode a test face in NMF?	H stays fixed and we solve a nonnegative least-squares problem for each face's weights. An exact solver gives the same test MSE.
What does the interval mean?	It is a paired bootstrap over the 80 test images (10,000 resamples). It does not cover other splits or other people.